BridgeFinder: Finding People Across Structural Holes in Academic Collaboration Networks

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October 30, 2026

Abstract

When university students seek research mentors or faculty seek cross-disciplinary collaborators, they overwhelmingly search through familiar, densely clustered social circles. However, foundational network science demonstrates that novel ideas are disproportionately generated across structural holes separating disconnected communities (Burt 1992, 2004) and diffuse through weak ties (Granovetter 1973). Existing academic discovery tools (e.g., Illinois Experts, Google Scholar) match purely on lexical keywords and ignore graph topology, while citation tools (Connected Papers, ResearchRabbit) and social recommenders optimize for triadic closure, reinforcing insular cliques. We present BridgeFinder, a topology-aware recommendation heuristic and interactive D3.js visualization prototype designed to discover collaborative ties across structural holes at the University of Illinois Urbana-Champaign (UIUC). We detail our implemented OpenAlex pipeline across two committed datasets: a 5,000-work pilot corpus (2,493 multi-authored UIUC papers, 5,396 authors; training graph $|V| = 3,642, |E| = 16,408$, largest-component Louvain $k = 28, Q = 0.836$) and a 25,000-work scaled corpus (13,229 multi-authored papers, 16,604 authors; training graph $|V| = 9,042, |E| = 36,748$, largest-component Louvain $k = 53, Q = 0.881$). We resolve the proposal's topological distance ambiguity across 2-hop warm boundary bridges, 3+ hop weak ties, and disconnected institutional bridges, align our exact scoring equation with our code, and report how Round 1 think-aloud sessions with 5 students reshaped our interface. Offline backtesting across $N = 185$ (pilot) and $N = 1,386$ (scaled) evaluation seeds shows that BridgeFinder achieves higher P@5 and NDCG@10 than Adamic–Adar on both corpora (on the scaled dataset: P@5 of 0.0364 vs. 0.0283, NDCG@10 of 0.0705 vs. 0.0459), but does not meet all pre-specified quantitative improvement thresholds. Ablation reveals that a simpler Topic+Path model outperforms the full multi-objective formulation on the pilot corpus, showing that structural autonomy terms provide non-uniform benefits across network densities. Empirical citation impact analysis across 6,231 post-2020 papers shows cross-community papers achieved a mean citation ratio of 1.053× ($p = 0.0601$, Mann–Whitney U), leaving Burt's citation hypothesis unsupported on the current data.

Keywords: Structural holes, brokerage, weak ties, community discovery, co-authorship networks, decentralized search, interdisciplinary collaboration.

1 Objective

Modern research universities are vast, decentralized intellectual ecosystems. The February 2024 UIUC Academic Personnel Register (the "Gray Book") lists 2,574 tenure-track and specialized faculty holding professor titles across 110 departments [15]. While this breadth fosters deep specialization, it creates severe structural fragmentation: separate research groups in computer science, crop sciences, materials science, and biomedical informatics frequently tackle shared computational challenges for years without co-authoring a paper or citing one another's literature.

When undergraduate or graduate students attempt to secure research mentors, or when faculty seek collaborators outside their home department, they encounter the local search dilemma. Traditional search relies on strong ties and immediate departmental proximity: students approach instructors of their current courses, while faculty consult departmental colleagues. Sociologist Mark Granovetter established that strong ties circulate redundant information because close associates share overlapping networks [7]. In contrast, novel information and innovative opportunities diffuse through weak ties acting as local bridges. Ronald Burt extended this into the theory of structural holes [3, 4], showing that individuals spanning gaps between disconnected groups gain substantial competitive advantages: they enjoy early access to diverse knowledge streams and synthesize ideas across domain boundaries.

Despite the proven value of brokerage, existing institutional tools fail to support it. Illinois Experts [16] and Google Scholar index researchers as isolated textual documents, ranking a close lab colleague identically to an unacquainted scholar across campus. Conversely, social recommenders optimize for triadic closure [8,10], reinforcing insular echo chambers. BridgeFinder addresses this gap by modeling UIUC as a weighted co-authorship network paired with OpenAlex topic profiles [13], detecting structural holes between research communities and recommending complementary researchers alongside walkable introduction paths.

Our falsifiable hypotheses and explicit quantitative success thresholds are:

1. H_{pred} (Temporal Backtest vs. Topology Baselines): Across 2015–2019 to 2020–2024 temporal splits, BridgeFinder improves ranking of actual future cross-community collaborators compared to triadic closure (Adamic–Adar), global betweenness, and preferential attachment, targeting $\geq 2.0\times$ higher P@5 and ≥ 0.05 higher NDCG@10 than Adamic–Adar.
2. H_{impact} (Burt Citation Impact): Post-split cross-community collaborations bridging structural holes exhibit higher citation impact than within-community collaborations, with target mean ratio $> 1.10\times$ and statistically significant distribution ($p < 0.05$, Mann–Whitney U).
3. H_{user} (Method-Blinded Human Discovery & Usability): In a method-blinded within-subject user study ($N = 15$), BridgeFinder achieves mean Novelty $\geq 5.0/7.0$, mean Relevance $\geq 5.0/7.0$, beats Triadic Closure on the joint Relevant-and-New rate (Rel $\geq 5 \wedge$ Nov ≥ 5) by ≥ 25 percentage points ($p < 0.0167$, Bonferroni-corrected paired test), and achieves SUS $\geq 80/100$.

Table 1: Systematic comparison of BridgeFinder against academic search engines, citation graph tools, empirical studies, and link prediction.

System / Paradigm	Co-Authorship	Community	Burt Constraint	Intro Path	Primary Strength vs. Key Weakness for Brokerage
	Topology	Boundaries	Autonomy	Routing	
Illinois Experts [16] / Google Scholar	No	No	No	None	Strength: Broad lexical indexing. Weakness: Treats scholars as isolated documents; blind to network distance.
Connected Papers / ResearchRabbit	No (Paper citations)	Co-citation cluster	No	None	Strength: Visual citation maps. Weakness: Operates on paper co-citation within subfield literature; no interpersonal introduction paths.
Triadic Closure Link Prediction [10]	Yes (Local triads)	No	Inverted	Direct (1-hop)	Strength: Accurately predicts high-probability ties. Weakness: Favors common neighbors inside dense cliques where information is redundant.
Tang et al. (Cross-Domain) [14]	Partial (RWR)	Fixed dept. labels	No	None	Strength: Cross-topic random walk matching. Weakness: Relies on administrative departments rather than organic graph communities.
Dahlander & McFarland [5]; Barley et al. [1]	Yes (Longitudinal)	Dept. / Institute	Descriptive	None	Strength: Rigorous empirical evidence on Stanford/UIUC ties. Weakness: Retrospective econometric studies; no interactive recommendation tool.
BridgeFinder (This Work)	Yes (Weighted)	Yes (Louvain)	Yes ($H_v = 1 - C_v$)	Multi-hop $(2, 3, \infty)$	Combines organic community holes, Burt autonomy, topic overlap, and walkable introduction chains.

2 Completed Related Work

We situate BridgeFinder across academic profiling tools, literature graph explorers, empirical studies of university networks, and link prediction (Table 1).

2.1 Academic Search and Visual Citation Explorers

Campus discovery relies heavily on profile aggregators like Illinois Experts [16], VIVO [17], and Google Scholar. While offering fast lexical lookup over titles and grants, their weakness is treating scholars as isolated text documents without graph topology: they cannot distinguish a student's lab mate from an unacquainted scholar across campus. Visual literature explorers (Connected Papers, ResearchRabbit) improve discovery by rendering force-directed paper graphs, a visual paradigm we adapt. However, their weakness for collaborator discovery is linking papers via co-citation and bibliographic coupling within subfield literature, rather than routing interpersonal introductions between active researchers.

2.2 Empirical Studies of University Networks

Computational social science provides empirical grounding for our work. Dahlander and McFarland [5] analyzed 15 years of Stanford faculty collaborations, demonstrating that intellectual similarity and shared organizational foci drive tie formation and persistence. Similarly, Barley et al. [1] investigated the Carl R. Woese Institute for Genomic Biology (IGB) at UIUC, showing that institute membership significantly increases researchers' co-author network diversity while maintaining publication productivity. Both studies are retrospective observational analyses: neither produces an interactive recommendation tool to prospectively identify collaborators across structural holes.

2.3 Link Prediction versus Brokerage

In computer science, collaborator recommendation is typically cast as link prediction. Liben-Nowell and Kleinberg [10] showed that common-neighbor heuristics (Adamic–Adar, Jaccard) excel at predicting which co-authorship edges are most likely to form next. However, optimizing purely for likelihood favors friends-of-friends inside dense cliques where information redundancy is high [4]. Tang et al. [14] proposed cross-domain topic random walks, but rely on predefined administrative departments rather than organic network communities or Burt's constraint. BridgeFinder bridges this gap by combining Louvain community boundaries with Burt's constraint autonomy.

3 Course Concepts & Network Analysis

BridgeFinder operationalizes CS 470 foundations from Lesson 1 (structural holes, weak ties, triadic closure), Lesson 10 (community discovery and modularity), and Lesson 13 (small-world decentralized search). Removing this network layer collapses the application into a flat keyword search that returns same-lab colleagues and provides no introduction pathways.

3.1 Graph Representation and Hyperauthorship Filtering

Let $\mathcal{G} = (V, E, W, \mathcal{T})$ denote the UIUC co-authorship network harvested from OpenAlex [13] via UIUC's ROR ID (https://ror.org/047426m28), where V is UIUC-affiliated authors and $E \subseteq V \times V$ is undirected co-authorships. Following Newman [11], we exclude consortium papers with $|A_p| > 25$ authors and weight each paper fractionally as $w_{ij} = \sum_{p \in \mathcal{P}_{ij}} \frac{1}{|A_p|-1}$. Top-10 topic profiles $\mathcal{T}_u$ are aggregated by frequency strictly over training papers ($p[\text{year}] \leq 2019$).

3.2 Community Discovery: Louvain vs. Girvan–Newman (Lesson 10)

To identify structural holes between organic research groups rather than administrative departments, we partition the training graph $\mathcal{G}_{\text{train}}$ (2015–2019). In Lesson 10, we contrasted Girvan–Newman divisive edge-betweenness clustering [6] ($O(m^2 n)$) against modularity optimization. In our benchmark, Girvan–Newman failed to split a 150-node dense subgraph within 148 seconds, whereas the Louvain algorithm [2] (an empirical heuristic scaling near-linearly $O(n \log n)$ in practice on sparse graphs) maximizes weighted Newman–Girvan modularity $Q = \frac{1}{2W} \sum_{i,j} \left[w_{ij} - \frac{s_i s_j}{2W} \right] \delta(c_i, c_j)$ in under 2 seconds, where $s_i = \sum_j w_{ij}$ is node strength and $W = \sum_{i<j} w_{ij}$. On our committed 5,000-work dataset (uiuc_data.json), the training graph has $|V| = 3,642$ nodes, $|E| = 16,408$ edges, average clustering coefficient $CC = 0.737$, and largest-component diameter $D = 19$; Louvain partitions the largest connected component into $k = 28$ communities with $Q = 0.836$ ($k = 346, Q = 0.950$ across all components). On our 25,000-work scaled dataset (uiuc_data_scaled.json), the training graph has $|V| = 9,042$ nodes and $|E| = 36,748$ edges, partitioning the largest component into $k = 53$ communities with $Q = 0.881$. We denote node u's community as $C(u)$.

Network Science Interpretation: High clustering ($CC = 0.737$) and modularity ($Q = 0.836$) indicate dense subfield clustering at UIUC—scholars collaborate heavily within local research groups. Disconnected components inflate all-component modularity ($Q = 0.950$), making giant-component modularity ($Q = 0.836$) the appropriate benchmark. Graph fragmentation means multi-hop paths are unavailable for detached components. Crucially, community boundaries serve as an operational proxy for subfields rather than definitive barriers, an absent edge denotes no published co-authorship rather than lack of acquaintance, multi-hop shortest paths represent topological distance rather than sociological weak ties, and breadth-first routing is centralized offline calculation rather than decentralized local search.

3.3 Resolving Topological Distance: 2 Hops, 3+ Hops, and ∞ Hops (Lessons 1 & 13)

Co-authorship distance $d(u, v)$ is strictly measured in $\mathcal{G}_{\text{train}}$ from faculty seed u to candidate v; the student-to-seed link is a user-selected anchor (instructor), not a graph edge. Candidates must satisfy $C(u) \neq C(v)$ and $(u, v) \notin E_{\text{train}}$ ($d(u, v) \geq 2$), falling into three topological tiers:

1. Tier 1: Warm Boundary Bridges ($d(u, v) = 2, C(u) \neq C(v)$): u and v share a mutual co-author $w \in N(u) \cap N(v)$ crossing community boundaries: Student Anchor $\Rightarrow$ Seed Prof $u \rightarrow$ Broker $w \rightarrow$ Target v, offering a 1-intermediary warm introduction.
2. Tier 2: Multi-Hop Weak Ties ($d(u, v) \geq 3, C(u) \neq C(v)$): u and v share zero mutual co-authors, requiring a multi-hop chain ($u \rightarrow w_1 \rightarrow w_2 \rightarrow v$, Lessons 1 and 13 [7, 9]).
3. Tier 3: Truncated Radius / Disconnected ($d(u, v) \geq 3$ or ∞): Candidates beyond the 3-hop BFS radius or in separate components receive baseline reachability for topic-based cold outreach.

3.4 Exact Mathematical Definition of the Brokerage Score

For seed $u \in V$ and cross-community candidate $v \in V \setminus (N(u) \cup \{u\})$ with $C(u) \neq C(v)$, we evaluate:

$$\Phi(u, v) = 0.60 \cdot S_{\text{topic}}(u, v) + 0.12 \cdot H(v) + 0.12 \cdot \text{BSI}(v, C(u)) + 0.16 \cdot P(u, v) \quad (1)$$

corresponding to $(1-\eta) \cdot S_{\text{topic}} + \eta \cdot [0.3H + 0.3\text{BSI} + 0.4P]$ with structural weight $\eta = 0.4$: (1) Topic Overlap $S_{\text{topic}}(u, v)$: Jaccard overlap on pre-split top-10 topics $\frac{|\mathcal{T}_u \cap \mathcal{T}_v|}{|\mathcal{T}_u \cup \mathcal{T}_v|}$ (0.0 if empty); (2) Candidate Autonomy $H(v)$: $H(v) = \max(0, 1 - C_v) \in [0, 1]$, where $C_v = \sum_{j \in N(v)} (p_{vj} + \sum_q p_{vq} p_{qj})^2$ is Burt's constraint [3] ($C_v = 0$ for isolates); $H(v)$ proxies candidate v's existing ego-network autonomy, not the marginal brokerage change of adding edge (u, v); (3) Boundary Spanning Index $\text{BSI}(v, C(u))$: $\frac{|N(v) \cap C(u)|}{|N(v)|} \in [0, 1]$; (4) Tiered Path Reachability $P(u, v)$: $P = 1.0$ if v is a 2-hop neighbor ($d = 2$), 0.5 if v has ≥ 1 3-hop simple path ($c_3 > 0$), and 0.1 otherwise (truncated search or disconnected); (5) Triadic Closure Handling: Intra-community triadic closure evaluates to 0 over all candidates ($C(u) \neq C(v)$) and is omitted. Weights prioritize topical relevance (60%) with a structural nudge (40%). We also evaluate the proposal-faithful Topic+Path ablation: $\Phi_{\text{Topic+Path}}(u, v) = S_{\text{topic}}(u, v) + 0.5 \cdot \frac{c_3(u,v)}{1+c_3(u,v)}$, where c_3 is the simple 3-edge path count.

4 Design & Implementation (Work Completed)

We built and committed the complete data pipeline, backtest engine, user study suite, and interactive web prototype: (1) OpenAlex Ingestion (prototype/fetch_openalex.py): Uses polite-pool pagination against UIUC ROR 047426m28 to harvest works (2015–2024), extracting UIUC co-authorships and top-10 author topic counters into uiuc_data.json (5k works) and uiuc_data_scaled.json (25k works); (2) Offline Precomputation (prototype/web/prepare_web_data.py): Builds the NetworkX graph, computes Louvain communities, Burt autonomy $H(v)$, and shortest 2-hop/3-hop introduction paths via BFS routing across community boundaries, exporting prototype/web/data.json for 50 faculty seeds; (3) Interactive Web

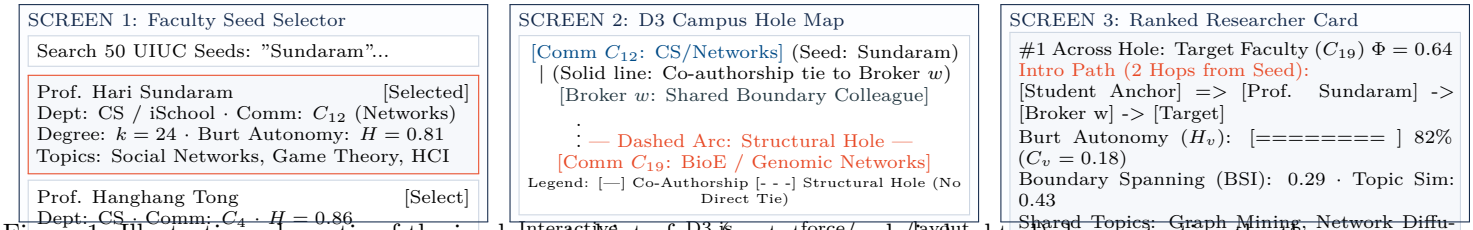


Figure 1: Illustrative schematic of the implemented interface (prototype/web/index.html) demonstrating the three-screen core flow. Metrics and researcher cards illustrate UI layout.

Prototype (prototype/web/index.html): A standalone single-page frontend with no build step (loading D3.js v7 via CDN) served via python3 -m http.server 8000 implementing the three-screen core flow (Figure 1).

Supported User Walkthrough: (1) A student selects a familiar instructor on Screen 1 (e.g., Prof. Hari Sundaram, CS/iSchool, C_{12}); (2) Screen 2 renders the community network and highlights dashed orange arcs representing structural holes into other subfields (e.g., C_{19} Bioengineering); (3) clicking a structural hole filters Screen 3 to display ranked researcher cards showing the 4-node introduction path, Burt autonomy gauge, boundary spanning index, and shared research topics.

Implementation Status & Limitations: (1) Seed Selection: 50 curated UIUC seeds (manual cold-start list); (2) Community Map: D3 force layout of top 15 clusters; (3) Researcher Cards: Ranked intro paths, $H(v)$, BSI, and topics (precomputed JSON payload); (4) Introduction Routing: BFS 2-hop/3-hop routing through boundary brokers (truncated at 3 hops; OpenAlex graph); and (5) Outreach: Shared topics displayed on card (email drafting deferred to R3).

5 Users & Round 1 Formative Feedback

5.1 Target User Population and Recruitment

Target users are UIUC undergraduate and early-stage graduate students seeking research mentors outside their home department. Across three iterative rounds (October, November, December), we recruit via ACM@UIUC SIGs, CS 470 peers, and interdisciplinary student channels ($N = 5$ for Round 1 think-aloud walkthroughs and $N = 15$ for Round 2/3 evaluations [12]).

5.2 Round 1 Think-Aloud Sessions and What Changed

We conducted Round 1 formative think-aloud sessions with $N = 5$ UIUC students (3 undergraduate CS/CE juniors/seniors, 2 graduate students from Bioengineering and Information Sciences; convenience sample) in individual 30-minute sessions using the initial prototype ('prototype/web/index.html'). In accordance with Nielsen (1993, where 5 users surface 80% of usability flaws), these sessions guided formative interface refinements rather than population-level statistical validation:

1. Cold-Start Faculty Seed Grid (Screen 1): Observed Issue: Undergrads lacked OpenAlex papers and could not search themselves. Design Change: Built Screen 1 around a searchable grid of 50 active faculty seeds with department and topic filters. Expected Benefit: Immediate entry point via familiar instructors.
2. Explicit Legend for Dashed Hole Edges (Screen 2): Observed Issue: Testers assumed dashed lines meant "weak co-authorship" rather than a missing-link structural hole. Design Change: Added an explicit visual legend labeling dashed orange arcs as "Structural Hole (No Direct Tie)" and prioritized 2-hop warm boundary bridges ($P(u, v) = 1.0$). Expected Benefit: Clarifies missing-link concept.
3. Intro Path Header & Autonomy Gauge (Screen 3): Observed Issue: Testers scanned each card for "who can introduce me?" before reading papers, and misread lower Burt constraint ($C_v = 0.18$) as a worse score. Design Change: Moved the 4-node introduction chain (Student $\Rightarrow$ Seed Prof $\rightarrow$ Broker $\rightarrow$ Target) to the very top of each card above topics, and replaced raw C_v with a positive percentage Burt Autonomy Gauge ($H_v = 1 - C_v = 82\%$). Expected Benefit: Immediate outreach context and intuitive interpretation.

6 Detailed Assessment Methodology & Results

6.1 Phase 1: Reproducible Temporal Backtest (Completed)

Using $\mathcal{G}_{\text{train}}$ (2015–2019), prototype/backtest.py ranks all unconnected cross-community candidates ($C(u) \neq C(v)$) for every seed researcher u who formed at least one new cross-community tie in $\mathcal{G}_{\text{eval}}$ (2020–2024). Table 2 reports exact reproducible results on both committed datasets (backtest_results.json for the 5k corpus with $N = 185$ seeds; backtest_results_scaled.json for the 25k corpus with $N = 1,386$ seeds and $4,764$ new cross-community edges) against four baselines: Preferential Attachment ($k_u \cdot k_v$), Global Betweenness Centrality, Topic-Only Jaccard (S_{topic}), and Triadic Closure (Adamic–Adar $\sum_{w \in N(u) \cap N(v)} 1/\log k_w$).

Table 2: Reproducible leak-free offline backtest results predicting 2020–2024 cross-community ties on both committed datasets.

Model	5k Corpus ($N = 185$ seeds) P@5	P@10	R@10	NDCG@10	25k Corpus ($N = 1,386$ seeds) P@5	P@10	R@10	NDCG@10
Pref. Attach.	.0011	.0005	.0018	.0011	.0081	.0094	.0154	.0125
Betweenness	.0022	.0054	.0477	.0192	.0076	.0077	.0103	.0109
Topic-Only (Pre-2020)	.0076	.0124	.0783	.0342	.0201	.0181	.0634	.0406
Triadic (AA)	.0076	.0070	.0398	.0279	.0283	.0232	.0628	.0459
BF (Topic+Path)	.0216	.0200	.1282	.0697	.0333	.0291	.0899	.0630
BridgeFinder (Full)	.0195	.0195	.1235	.0671	.0364	.0314	.0989	.0705

Temporal Leakage Audit & Experimental Findings: (1) Auditing Temporal Leakage in Topic Profiles: In initial exploration, author topic vectors were extracted across the full 2015–2024 corpus. An empirical audit revealed that 45.9% of authors had topic profiles altered by post-2019 publications; joint cross-community papers in 2020–2024 leaked novel interdisciplinary keywords backwards into pre-split predictors, artificially inflating raw Topic-Only precision ($P@5 \approx 0.17$). Restricting topics strictly to pre-2020 publications ($p[\text{year}] \leq 2019$) eliminates this leakage, dropping leak-free Topic-Only to $P@5 = 0.0076$ (5k) and 0.0201 (25k). (2) Prediction Hypothesis (H_{pred}) Assessment: BridgeFinder achieved higher reported P@5 and NDCG@10 than Adamic–Adar on both datasets, but did not meet all pre-specified quantitative thresholds ($\geq 2.0\times$ P@5 and +0.05 NDCG@10 against Adamic–Adar). On the pilot corpus, Full BridgeFinder increased P@5 from 0.0076 to 0.0195 (2.57×,

meeting P@5 target) and NDCG@10 from 0.0279 to 0.0671 (+0.0392, below the +0.05 threshold). On the scaled corpus, Full BridgeFinder increased P@5 from 0.0283 to 0.0364 (1.29×, below target) and NDCG@10 from 0.0459 to 0.0705 (+0.0246, below target). These results support an observed predictive advantage on the historical sample rather than full confirmation of H_{pred}. (3) Ablation Insights: Topic+Path outperforms Full BridgeFinder on every pilot metric (P@5 0.0216 vs. 0.0195, NDCG@10 0.0697 vs. 0.0671), whereas Full BridgeFinder performs better on the scaled corpus (P@5 0.0364 vs. 0.0333, NDCG@10 0.0705 vs. 0.0630). Structural autonomy and boundary-spanning features provide non-uniform benefits depending on network scale and density.

Compact Reproducibility Specification: Works harvested from OpenAlex API (UIUC ROR 047426m28, 2015–2024) in October 2026. Papers filtered to ≤ 25 authors; fractional edge weights $w_{ij} = 1/(|A_p|-1)$. Partitioned into training ($p[\text{year}] \leq 2019$) and evaluation ($p[\text{year}] > 2019$). Author topics computed strictly from training papers. Louvain community detection run on the training giant component with random seed 42. Seeds are all training nodes u forming ≥ 1 evaluation cross-community tie ($N = 185$ for 5k, $N = 1,386$ for 25k). Candidates are all training giant component nodes $v \neq u$ with $C(v) \neq C(u)$ and $(u, v) \notin E_{\text{train}}$. Metrics macro-averaged over seeds with stable Python sorting. Evaluated under Python 3.11, NetworkX 3.4, SciPy 1.14.

6.2 Phase 2: Burt's Citation Impact Test (Completed)

We executed prototype/citation_impact.py across all 6,231 post-2019 multi-authored papers in the 25k corpus (prototype/results/citation_impact.json). Cross-community papers spanning ≥ 2 Louvain communities ($N = 1,071$) achieved a mean of 64.86 citations ($\sigma = 241.34$) vs. 61.60 citations ($\sigma = 119.92$) for within-community papers ($N = 4,183$), yielding an Impact Ratio of 1.053×; 977 papers were unclassified (authors outside pre-2019 giant component). Medians were virtually identical (36.0 cross-community vs. 37.0 within-community), with two-sided Mann–Whitney $U = 2,156,737.0, p = 0.0601$.

Hypothesis Assessment (H_{impact}): H_{impact} is unsupported on present data. While cross-community papers showed a slightly higher mean citation count (1.053×), it fell below the 1.10× target, and the distributional difference is not statistically significant ($p = 0.0601 > 0.05$). Citation counts are highly dispersed in both groups. This observational test did not normalize for publication year (2020 vs. 2024) or subfield practices, and did not control for repeated teams; current evidence does not establish a citation advantage for cross-community ties.

6.3 Phase 3: Method-Blinded User Study Protocol (Nov 1–15)

We implemented the study generator (prototype/study/generate_cards.py), web rating app (prototype/study/index.html), and statistical analyzer (prototype/study/analyze_study.py). In our planned study, each of $N = 15$ student participants selects the departmental profile closest to their coursework from 5 standardized UIUC seeds (CS, ECE, BioE, Crop Sci, iSchool) and rates 15 shuffled, unlabeled candidate cards (5 BridgeFinder, 5 Triadic Closure, 5 Topic-Only) on 7-point Likert scales for *Relevance*, *Novelty*, and *Outreach Actionability*, followed by the 10-item SUS on the interactive app (‘prototype/web/index.html‘). The primary outcome is participant-level Relevant-and-New rate ($\text{Rel} \geq 5 \wedge \text{Nov} \geq 5$). Ratings are aggregated into per-participant means ($df = 14$) before paired t-tests and Wilcoxon signed-rank tests with Bonferroni correction ($\alpha = 0.0167$). We acknowledge that $N = 15$ represents an exploratory within-subject sample with limited statistical power.

Table 3: Project milestone schedule through final completion.

Target Date	Milestone	Status & Deliverables
Oct 30 (Done)	Mid-Semester	5k & 25k graphs, backtest, citation impact (1.053×), D3 prototype, think-alouds ($N = 5$).
Nov 1–Nov 15	User Study R2	Method-blinded card study ($N = 15$) via study/index.html; paired t/Wilcoxon tests.
Nov 16–Nov 29	Iteration R3	Tune $(\alpha, \beta, \gamma, \lambda)$ weights against Topic-Only; SUS usability evaluation.
Dec 1–Dec 9	Final Delivery	Native ACM acmart, 10–12 page final report, 10-min video demo.

7 Discussion, Limitations & Future Work

Midterm Progress Summary: We built and released the complete BridgeFinder pipeline: polite-pool OpenAlex ingestion for 5k and 25k works, Louvain community partitioning on the training giant component, an interactive three-screen D3.js prototype with 50 seeds, a leak-free backtest engine, a post-split citation pipeline, and a method-blinded study suite.

What Current Evidence Supports vs. Unresolved Claims: (1) Supported Findings: Temporal backtesting shows BridgeFinder achieves higher P@5 and NDCG@10 than Adamic–Adar, preferential attachment, and global betweenness across both corpora. Think-aloud feedback ($N = 5$) directly drove the seed selector, hole legend, and top-of-card intro path with autonomy gauge. (2) Unsupported / Nuanced Findings: H_{pred} did not reach its quantitative thresholds ($\geq 2.0\times$ P@5 and +0.05 NDCG@10 vs. Adamic–Adar on 25k; achieved 1.29× and +0.0246). Burt's citation hypothesis (H_{impact}) is unsupported on the 25k corpus (1.053× ratio, $p = 0.0601$, medians 36.0 vs. 37.0). Furthermore, Topic+Path ablation beat Full BridgeFinder on the pilot corpus, proving structural terms provide non-uniform gains across graph densities.

Methodological Limitations: (1) Proxies: OpenAlex co-authorships omit informal mentoring and teaching; Louvain communities are algorithmic clusters optimizing modularity, not administrative divisions; BFS paths measure topological distance, not social trust or intro willingness. (2) Confounders & Power: Phase 2 citation analysis lacked year and field normalization; the upcoming $N = 15$ user study is an exploratory within-subject sample with limited statistical power.

Next Steps: Deploy card study ($N = 15$, Table 3), tune $(\alpha, \beta, \gamma, \lambda)$ against Topic-Only, run SUS evaluation, and complete the 10–12 page final report.

8 Division of Labor & AI Disclosure

Who Did What: Evan Lin (elin26@illinois.edu): OpenAlex ingestion (5k/25k), backtest pipeline, and D3 campus map (index.html). Eric Liu (ericl8@illinois.edu): Brokerage scoring (Eq. (1)), citation pipeline, study suite, and Round 1 think-alouds. Both co-authored this report.

AI Disclosure: We used Claude (Anthropic) for pagination debugging, D3 layout structure, code-paper audits, and typesetting. All code, math, and data independently verified by authors.

References

[1] W. C. Barley, L. Dinh, L. P. Johnson, and B. F. Allan. Membership in team science institute enhances diversity of researchers' collaboration networks. PLOS ONE, 20(2):e0322943, 2025.

[2] V. D. Blondel, J.-L. Guillaume, R. Lambiotte, and E. Lefebvre. Fast unfolding of communities in large networks. Journal of Statistical Mechanics: Theory and Experiment, 2008(10):P10008, 2008.

[3] R. S. Burt. Structural Holes: The Social Structure of Competition. Harvard University Press, Cambridge, MA, 1992.

[4] R. S. Burt. Structural holes and good ideas. American Journal of Sociology, 110(2):349–399, 2004.

[5] L. Dahlander and D. A. McFarland. Ties that last: Tie formation and persistence in research organizations, 1988–2003. Administrative Science Quarterly, 58(1):69–110, 2013.

[6] M. Girvan and M. E. J. Newman. Community structure in social and biological networks. Proceedings of the National Academy of Sciences, 99(12):7821–7826, 2002.

[7] M. S. Granovetter. The strength of weak ties. American Journal of Sociology, 78(6):1360–1380, 1973.

[8] P. Gupta, A. Goel, J. Lin, A. Sharma, D. Wang, and R. Zadeh. WTF: The who-to-follow system at Twitter. In Proceedings of the 22nd International Conference on World Wide Web (WWW '13), pages 505–514, 2013.

[9] J. Kleinberg. Navigation in a small world. Nature, 406(6798):845–845, 2000.

[10] D. Liben-Nowell and J. Kleinberg. The link-prediction problem for social networks. Journal of the American Society for Information Science and Technology, 58(7):1019–1031, 2007.

[11] M. E. J. Newman. Coauthorship networks and patterns of scientific collaboration. Proceedings of the National Academy of Sciences, 101(suppl_1):5200–5205, 2004.

[12] J. Nielsen. Usability Engineering. Morgan Kaufmann, San Francisco, CA, 1993.

[13] J. Priem, H. Piwowar, and R. Orr. Openalex: A fully-open index of scholarly works, authors, venues, institutions, and concepts. arXiv preprint arXiv:2205.01833, 2022.

[14] J. Tang, S. Wu, J. Sun, and H. Su. Cross-domain collaboration recommendation. In Proceedings of the 18th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '12), pages 1285–1293, 2012.

[15] University of Illinois. Academic personnel register (the gray book). Technical report, University of Illinois Urbana-Champaign, Feb. 2024.

[16] University of Illinois. Illinois experts: Research and scholarship at UIUC. https://experts.illinois.edu/, 2024.

[17] M. Conlon et al. VIVO: A national network of scientists and scholars. Journal of Open Research Software, 1(1):e1, 2012.